QONTOS Research Papers v2.1 - Executive Summary

Executive View

The QONTOS `v2.1` paper set now presents a more technically feasible research and production thesis than the earlier draft package.

The key change is structural:

- `current QONTOS` is presented as an implementation-grounded, software-first modular orchestration platform
- `future QONTOS` is presented as a staged research and engineering program aimed at modular hardware scale
- `2030 stretch outcomes` are treated as contingent targets rather than completed results

This makes the package substantially more credible for research, investor, engineering, and production planning use.

What The v2.1 Set Says

The rewritten package makes five core points:

1. QONTOS already has a real software and orchestration identity today.
2. Modular scaling remains the central system thesis.
3. Million-physical-qubit and `10,000`-logical-qubit outcomes are stretch targets, not current demonstrations.
4. Every major technical pillar now has explicit assumptions and validation gates.
5. The roadmap is now a gated program plan rather than a single-path promise.

Three-Layer Narrative

The package is strongest when read in three layers.

Layer 1: Implemented Today

This is the most credible current QONTOS story:

- circuit ingestion and normalization
- partitioning and scheduling
- simulator-backed execution
- result aggregation
- replayability and observability
- modular runtime concepts grounded in real software

Layer 2: Research-Backed Next

This is the technical research stack that must be validated to move beyond the current platform:

- scaled modular architecture
- tantalum-silicon qubit platform
- lower-overhead fault tolerance
- photonic interconnects
- AI-assisted decoding
- cryogenic and facility engineering
- benchmarking methodology

Layer 3: Stretch Roadmap

This is the long-range ambition layer:

- million physical qubits
- up to `10,000` logical qubits in the stretch case
- modular fault-tolerant systems
- FeMoco-class flagship workloads
- facility-scale cryogenic operation

Canonical Architecture Summary

The current working architecture constants are:

- `2,000` physical qubits per chiplet
- `5` chiplets per module
- `10,000` physical qubits per module
- `10` modules per system
- `100,000` physical qubits per system
- `10` systems per data center
- `1,000,000` physical qubits per data center

This architecture is now used consistently across the rewritten `v2` papers.

Scenario Framework

The package now uses three scenarios instead of one overcommitted future state.

Base

- early modular hardware demonstrations
- first logical qubits
- digital-twin and software-first scaling

Aggressive

- stronger multi-module operation
- improved QEC overhead
- medium-scale modular benchmarks

Stretch

- million physical qubits
- `10,000` logical qubits
- FeMoco-class flagship benchmark
- facility-scale deployment

The stretch case is only plausible if multiple assumptions land together across devices, QEC, decoding, photonics, and cryogenics.

Most Important Technical Corrections

The rewrite fixed several credibility problems from the previous package:

- architecture arithmetic is now synchronized across papers
- unsupported “10 billion operations within coherence” language was removed
- `100:1` overhead is now treated as a stretch target rather than a declared achieved result
- `<500 ns` decoding is now presented as a target latency budget
- FeMoco is now a stretch benchmark in a workload ladder, not a current-ready endpoint
- the roadmap now includes validation gates and fallback outcomes

How To Use The Papers

For engineering planning

Use:

- CANONICAL_ARCHITECTURE_CONSTANTS.md
- ASSUMPTIONS_REGISTER.md
- CLAIM_STATUS_LEDGER.md

For external communication

Lead with:

- the current software-first QONTOS platform
- the modular architecture thesis
- the benchmarking methodology
- the gated roadmap

Avoid presenting stretch targets as demonstrated facts.

For investors and partners

The correct framing is:

```
QONTOS has a real implementation-grounded platform today and a
research program designed to test whether modular quantum scale can
become technically and economically feasible over time.
```

Recommended Reading Order

1. README_v2.md
2. 01_QONTOS_Scaled_Architecture_v2.md
3. 09_Benchmarking_v2.md
4. 06_Software_Stack_v2.md
5. 03_Error_Correction_100to1_v2.md
6. 02_Tantalum_Silicon_Qubits_v2.md
7. 04_Photonic_Interconnects_v2.md
8. 05_AI_Decoding_v2.md
9. 07_Cryogenic_Infrastructure_v2.md
10. 08_Quantum_Algorithms_v2.md
11. 10_Roadmap_2030_v2.md

Bottom Line

The `v2.1` package is now much closer to something that can support a serious research and production conversation.

Its strongest technically honest message is:

- QONTOS is real today as a software-first modular platform
- the hardware scale story is a research program with explicit assumptions
- the stretch roadmap is ambitious but now framed as conditional engineering progress rather than guaranteed arrival

QONTOS Research Papers v2.1: Technically Feasible Research and Production Thesis

Overview

This collection contains the `v2.1` QONTOS research papers for the scaled modular architecture program. The set is now framed as a technically feasible research and production thesis rather than a flat declaration that every 2030 stretch metric has already been achieved.

The paper set should be read in three layers:

- `Implemented today`: the current software-first orchestration and digital-twin platform
- `Research-backed next`: modular architecture, qubit platform, QEC, photonics, cryogenics, and benchmarking methodology
- `Stretch roadmap`: million-physical-qubit and FeMoco-class flagship targets, contingent on specific validation gates

Claim Posture

Across the `v2.1` papers, major claims should be interpreted using these labels:

- `Demonstrated`
- `Simulated`
- `Derived from literature`
- `QONTOS target`
- `Stretch target`

The strongest public posture for the set is:

```
QONTOS has an implementation-grounded modular software platform today
and a research program aimed at aggressive and stretch modular
hardware milestones over time.
```

Canonical Architecture Framing

The source of truth for architecture constants is CANONICAL_ARCHITECTURE_CONSTANTS.md.

The canonical stretch architecture used across the rewritten set is:

- `2,000` physical qubits per chiplet
- `5` chiplets per module
- `10,000` physical qubits per module
- `10` modules per system

- `100,000` physical qubits per system
- `10` systems per data center
- `1,000,000` physical qubits per data center

Logical-qubit scale is scenario-based:

- `Base`: early logical qubits and modular demonstrations
- `Aggressive`: hundreds to low thousands of logical qubits
- `Stretch`: `10,000` logical qubits contingent on device, QEC, decoder, interconnect, and infrastructure validation

Reading Guide

1. Architecture

File: 01_QONTOS_Scaled_Architecture_v2.md

Purpose: - defines the canonical modular hierarchy - separates base, aggressive, and stretch scenarios - establishes validation gates and no-go conditions

2. Qubit Platform

File: 02_Tantalum_Silicon_Qubits_v2.md

Purpose: - evaluates the device platform needed for the architecture to be plausible - avoids unsupported “operations within coherence” arithmetic - focuses on coherence, fidelity, packaging, yield, and integration requirements

3. Error Correction

File: 03_Error_Correction_100to1_v2.md

Purpose: - frames `100:1` as a stretch target rather than a completed result - ties effective overhead to decoder quality, noise assumptions, and modular operation

4. Photonic Interconnects

File: 04_Photonic_Interconnects_v2.md

Purpose: - evaluates the interconnect path from literature baseline to stretch architecture - treats transduction efficiency and Bell-pair performance as gated targets

5. AI-Assisted Decoding

File: 05_AI_Decoding_v2.md

Purpose: - defines decoder latency budgets for base, aggressive, and stretch scenarios - treats sub-microsecond decoding as a target architecture question, not an achieved fact

6. Software Stack

File: 06_Software_Stack_v2.md

Purpose: - bridges the current implemented QONTOS platform to the future modular operating layer - clearly separates current software capability from future hardware-integrated capability

7. Cryogenic Infrastructure

File: 07_Cryogenic_Infrastructure_v2.md

Purpose: - treats cryogenics as a facility-scale planning and risk problem - links thermal feasibility to wiring, control density, redundancy, and operations

8. Algorithm Roadmap

File: 08_Quantum_Algorithms_v2.md

Purpose: - defines a benchmark ladder from small chemistry and optimization workloads to a FeMoco-class stretch target - avoids presenting FeMoco as current readiness

9. Benchmarking Methodology

File: 09_Benchmarking_v2.md

Purpose: - defines how QONTOS should measure progress - separates demonstrated metrics from target metrics - provides a credibility framework for the rest of the set

10. Roadmap

File: 10_Roadmap_2030_v2.md

Purpose: - translates the architecture into a gated program plan - introduces base, aggressive, and stretch outcomes rather than one promised path

Supporting Source-Of-Truth Documents

- TECHNICAL_FEASIBILITY_REWRITE_PACK.md
- CANONICAL_ARCHITECTURE_CONSTANTS.md
- ASSUMPTIONS_REGISTER.md
- CLAIM_STATUS_LEDGER.md

Recommended Public Positioning

The strongest technically honest public framing is:

- the current QONTOS platform is implementation-grounded and software-first
- the `v2.1` paper set defines the research program required to scale toward modular hardware systems

- the million-qubit and 10,000-logical-qubit figures are stretch targets, not current demonstrated results

Suggested Release Order

1. software and implemented platform story
2. architecture
3. benchmarking methodology
4. error correction
5. qubit platform
6. photonic interconnects
7. cryogenic infrastructure
8. algorithm roadmap
9. gated roadmap to 2030

Important Editorial Note

The rewritten v2.1 papers and the four source-of-truth support files remain the primary technical working set. Older non-v2 paper filenames and older paper-export filenames in this folder have now been synchronized as compatibility editions so they no longer contradict the v2.1 claim posture. Historical figures and older draft variants are still retained for reference.

QONTOS Scaled Quantum Architecture: A Scenario-Based Path to 1 Million Physical Qubits

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Advanced Quantum Systems Division, Abu Dhabi, UAE

Correspondence: research@zhyra.xyz | qontos@zhyra.xyz

Document Date: March 2026

Abstract

This paper outlines the target QONTOS scaled quantum architecture under explicit base, aggressive, and stretch scenarios. The stretch-2030 scenario aims for **1,000,000 physical qubits** and **10,000 logical qubits** through a four-tier hierarchy of chiplets, modules, systems, and data centers. The paper does not claim that this architecture has been demonstrated today. Instead, it defines the architecture constants, subsystem assumptions, and validation gates that would need to be satisfied for the stretch scenario to become technically credible. In particular, the paper treats **100:1 effective physical-to-logical overhead**, millisecond-class superconducting qubits, high-rate photonic interconnects, and large-scale cryogenic deployment as linked program assumptions rather than settled results.

Claim status: architecture target and stretch scenario, informed by literature and QONTOS internal planning

Keywords: quantum computing, modular architecture, fault-tolerant computing, distributed quantum systems, scenario planning

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1. Introduction

QONTOS is pursuing a modular quantum computing architecture in which superconducting chiplets are integrated into cryogenic modules, connected into systems, and ultimately deployed as facility-scale quantum data centers. The architectural thesis is that modularity can relax the scaling bottlenecks of monolithic quantum processors, provided that qubit quality, error correction, interconnect performance, and cryogenic infrastructure all improve together.

The purpose of this paper is not to present a fully demonstrated million-qubit machine. Its purpose is to define the architecture and the conditions under which such a machine could become feasible.

1.1 Why Modular Scale Matters

The path to large-scale useful quantum computation is constrained by:

- chip yield and frequency crowding
- thermal load and wiring density
- error-correction overhead
- inter-module communication fidelity
- classical decoder and control latency

Monolithic architectures face increasing stress on all five fronts as qubit count rises. QONTOS therefore treats modularity as a first-class design principle rather than a late-stage packaging choice.

1.2 Scope and Claim Status

This paper uses the following claim labels:

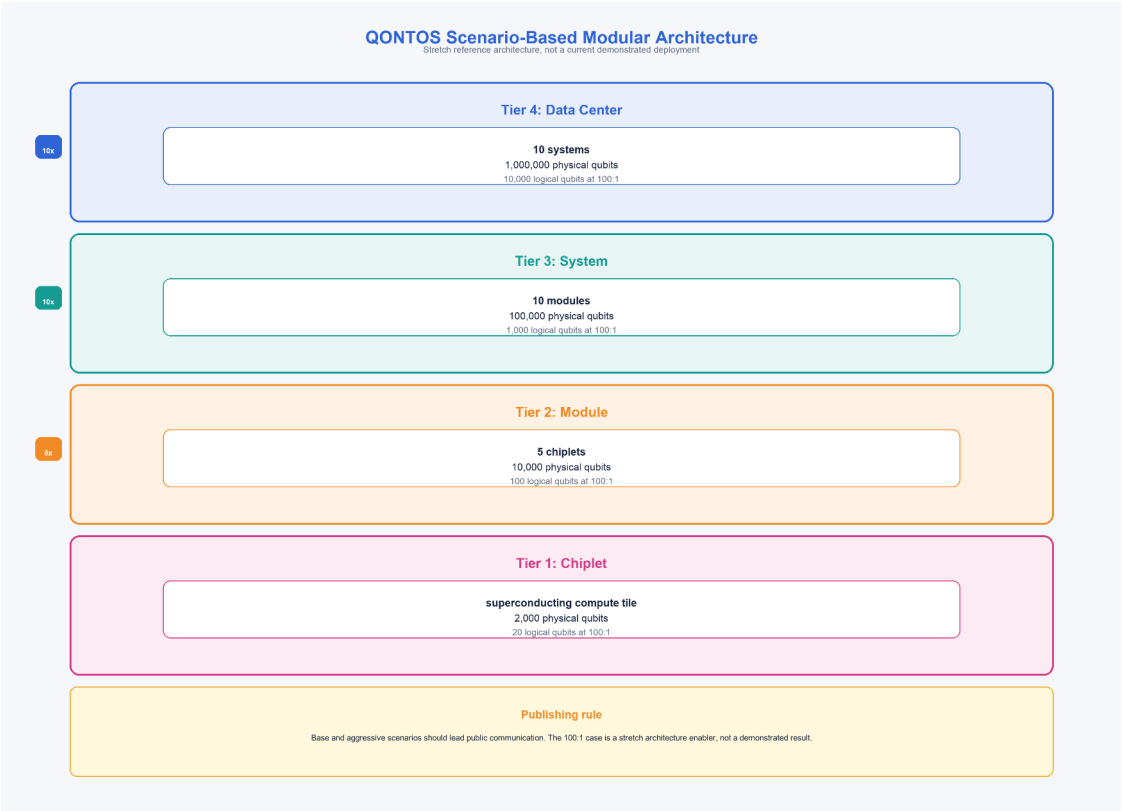
Label	Meaning
Demonstrated	Measured in QONTOS software or directly supported by published experimental results
Simulated	Supported by QONTOS digital-twin or simulator modeling
Derived from literature	Based on published external results or conservative extrapolation
QONTOS target	Internal engineering objective
Stretch target	Requires multiple major assumptions landing together

1.3 Architecture Claim Summary

Claim
QONTOS uses a four-tier modular hierarchy
2,000 physical qubits per chiplet
10,000 physical qubits per module
1,000,000 physical qubits by 2030
10,000 logical qubits by 2030
100:1 effective overhead

2. Canonical Architecture Constants

This section is the canonical source for architecture math across the research paper set.



Scaled Architecture

Figure 1: Target QONTOS four-tier hierarchy. The million-qubit configuration shown here is the stretch scenario, not a current demonstrated system.

2.1 Canonical Stretch Architecture

Layer
Chiplet
Module
System
Data center

2.2 Layer Definitions

Tier 1: QPU Chiplet

The chiplet is the fundamental fabricated superconducting qubit die.

Parameter	Stretch value	Notes
Physical qubits	2,000	Requires high-yield

		fabrication and manageable control density
Logical qubits at 100:1	20	Conditional on stretch QEC assumptions
Die size	15-20 mm	Placeholder target, not fabrication proof
Topology	Heavy-hex or equivalent sparse connectivity	Must remain compatible with control and error-correction assumptions

Tier 2: Quantum Module

The module is the cryogenic integration unit.

Parameter	Stretch value	Notes
Chiplets per module	5	Canonical constant
Physical qubits	10,000	Canonical constant
Logical qubits at 100:1	100	Stretch assumption
Inter-chiplet coupling	Microwave or equivalent short-range integration	Requires packaging and control validation

Tier 3: Quantum System

The system is the first photonic-connected machine layer.

Parameter	Stretch value	Notes
Modules per system	10	Canonical constant
Physical qubits	100,000	Canonical constant
Logical qubits at 100:1	1,000	Stretch assumption
Interconnect	Optical / photonic backbone	Requires transduction and entanglement validation

Tier 4: Quantum Data Center

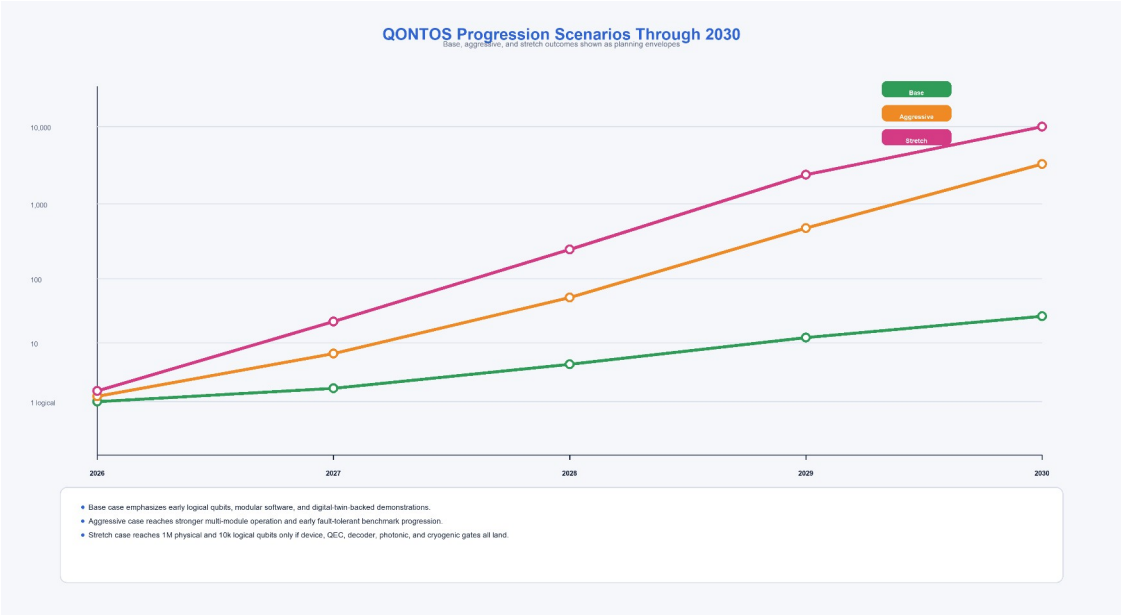
The data center is the facility-scale stretch endpoint.

Parameter
Systems per data center
Physical qubits
Logical qubits at 100:1
Power

Footprint

3. Scenario Framework

The million-qubit architecture should not be communicated as the only QONTOS future. This paper uses three scenarios.



Scenario Progression Timeline

Figure 2: Base, aggressive, and stretch scenario framing for QONTOS progression through 2030.

3.1 Base, Aggressive, and Stretch Cases

Metric
Physical qubits per chiplet
Physical qubits per module
Modules per system
Physical qubits per system
Systems per data center
Physical qubits per data center
Effective overhead
Logical qubits

3.2 Interpretation

- Base: early modular hardware plus strong software and digital-twin credibility

- Aggressive: large multi-module system with meaningful logical-qubit demonstrations
- Stretch: full million-qubit and 10,000-logical scenario

4. Error-Correction Envelope

4.1 Why 100:1 Is a Stretch Assumption

The transition from 1,000:1 to 100:1 effective overhead is central to the stretch architecture. However, this paper treats 100:1 as a contingent endpoint rather than a demonstrated property.

Overhead regime
1000:1
300:1
100:1

4.2 System Implications

For 10,000 logical qubits:

- At 1000:1 overhead: 10,000,000 physical qubits
- At 300:1 overhead: 3,000,000 physical qubits
- At 100:1 overhead: 1,000,000 physical qubits

This is why the million-qubit architecture and the 100:1 claim must always be presented together as a coupled stretch scenario.

4.3 Enabling Technologies

The stretch architecture depends on all of the following improving together:

1. millisecond-class qubits
2. high two-qubit fidelity at scale
3. low-overhead fault tolerance
4. fast decoder and control loops
5. photonic interconnect performance sufficient for modular operation

5. Roadmap Scenarios

5.1 Five Program Phases

Phase	Years	Base outcome	Aggressive outcome	Stretch outcome
FOUNDATION	2025-2026	improved software, digital	first small hardware	first evidence toward

		twin, early device validation	validation	millisecond-class devices
SPUTNIK	2026-2027	small modular hardware	scaled cryogenic module	10,000 physical qubits and early logical milestones
PIONEER	2027-2028	multi-module simulation and software maturity	hardware-linked distributed runtime	100,000 physical qubits and 1,000 logical qubits
HORIZON	2028-2029	robust modular control and benchmarks	meaningful FTQC demos	500,000 physical qubits and 5,000 logical qubits
SUMMIT	2029-2030	commercially useful sub-stretch system	large-scale logical platform	1,000,000 physical qubits and 10,000 logical qubits

5.2 Validation Gates

Gate	Why it matters
G1: device metrics cross agreed threshold	architecture cannot scale without qubit quality
G2: first logical qubits on modular substrate	roadmap cannot rely only on raw physical qubits
G3: inter-module entanglement and scheduling work in practice	architecture fails if modular links underperform
G4: digital twin and benchmark harness match experimental trajectory	roadmap becomes guesswork without this
G5: economics remain plausible under actual overhead and facility assumptions	million-qubit narrative fails if system cost explodes

5.3 No-Go Conditions

The stretch scenario should be downgraded if any of these occur:

- effective overhead remains closer to 1000:1 than 300:1
- photonic transduction stays in low-single-digit efficiency
- large chiplet packaging proves unmanageable
- cryogenic and control budgets fail to scale with module size

6. Performance Envelope

6.1 Scenario-Based Performance Table

Metric
Physical qubits
Logical qubits
Effective overhead
2Q fidelity
Transduction efficiency
Decoder latency

6.2 Competitive Framing

This paper should not present precise competitor 2030 numbers as settled facts. Instead:

- describe monolithic versus modular architectural differences
- compare scenario classes rather than exact future qubit totals
- reserve exact competitive comparisons for the benchmarking methodology paper

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7. Strategic Implications

If the stretch architecture becomes feasible, it would move QONTOS from an orchestration-layer company with a modular thesis to a full-stack modular quantum systems company. If the stretch architecture does not fully materialize, the aggressive and base scenarios still support a credible business in modular software, digital twin tooling, benchmarking, and early hardware integration.

This is why the scenario framing is important: it protects the core thesis even if the stretch case arrives later than hoped.

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8. Conclusion

QONTOS has a coherent modular scaling thesis, but the million-qubit architecture should be understood as a stretch scenario built on multiple linked assumptions. The value of this paper is therefore not that it proves million-qubit delivery today. Its value is that it defines a technically structured target architecture, a canonical system ladder, and a set of validation gates against which progress can be measured.

The most credible near-term message is:

- QONTOS has a software-first and modular-native foundation today
- QONTOS has an aggressive research roadmap toward large-scale modular quantum systems

- the million-qubit / 10,000-logical path remains a stretch target contingent on subsystem breakthroughs

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References

[1] S. Bland et al., “Two-millisecond coherence in superconducting transmon qubits via tantalum-on-silicon platform,” Nature, Nov. 2025.

[2] Google Quantum AI, “Suppressing quantum errors by scaling a surface code logical qubit,” Nature, 2023.

[3] S. Bravyi et al., “High-threshold and low-overhead fault-tolerant quantum memory,” Nature, 2024.

[4] IBM Research, “Quantum-centric supercomputing: The next wave in quantum computing,” 2024.

[5] McKinsey Global Institute, “Quantum Technology Monitor 2025,” 2025.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Architecture target with base/aggressive/stretch scenarios

Tantalum-on-Silicon Superconducting Qubits: Device Assumptions for a Modular Fault-Tolerant Path

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Correspondence: research@zhyra.xyz

Abstract

This paper examines the tantalum-on-silicon superconducting qubit platform as a candidate device basis for the QONTOS modular architecture. It does not claim that QONTOS has already demonstrated million-qubit device performance. Instead, it evaluates the device assumptions required for the aggressive and stretch architecture scenarios, including millisecond-class coherence, very high two-qubit fidelity, scalable chiplet fabrication, and manageable cryogenic and control integration. The paper treats literature-reported millisecond-class coherence as an encouraging input, then maps that input into realistic questions about logical depth, packaging, calibration, and fabrication yield.

Claim status: literature-informed device feasibility paper with aggressive and stretch targets

Keywords: superconducting qubits, transmon, tantalum, coherence, device roadmap, chiplet fabrication

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1. Introduction

The QONTOS architecture depends on a superconducting qubit platform that can simultaneously support:

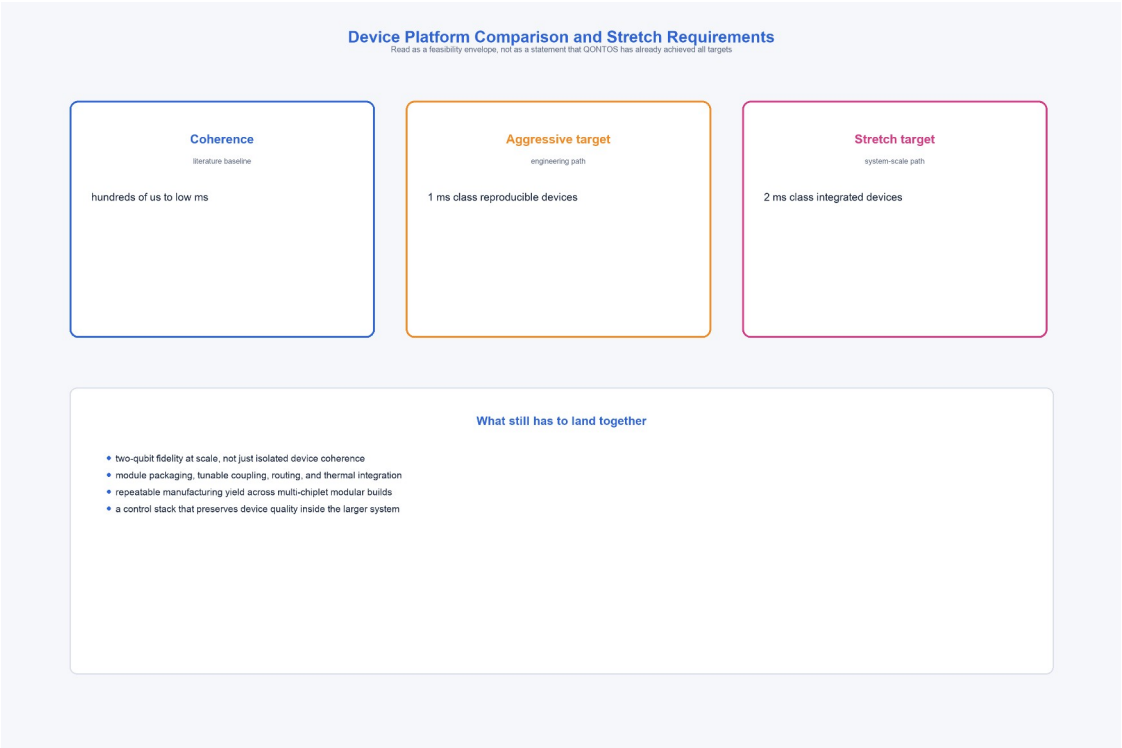
- high coherence
- high gate fidelity
- scalable fabrication
- modular packaging
- practical calibration and control

Tantalum-on-silicon is attractive because literature reports suggest significant progress over earlier material stacks. However, raw coherence values alone do not establish a viable path to fault-tolerant computation. This paper therefore focuses on what device-level performance would actually be needed for the larger architecture to remain credible.

1.1 Claim Status Summary

Claim	Status
Millisecond-class tantalum coherence exists in the literature	Derived from literature
QONTOS can build its stretch architecture on 2 ms devices	QONTOS target / stretch assumption
99.999% two-qubit fidelity at scale is available	Stretch target
Million-qubit fabrication is economically viable	Stretch target

2. Device Performance Requirements



Materials Comparison

Figure 1: Device-platform comparison should be read as a feasibility envelope, not as proof that stretch targets are already achieved.

2.1 Scenario-Based Targets

Metric
T1
T2
1Q fidelity

2Q fidelity

2Q gate time

Readout fidelity

2.2 Literature Baseline

The strongest role of tantalum in the current QONTOS paper set is as a literature-supported direction, not as an already integrated QONTOS device result.

Representative literature-relevant points:

- millisecond-class coherence has been reported for high-quality tantalum devices
- materials quality and oxide behavior appear favorable relative to older stacks
- device integration at scale remains far less mature than isolated coherence headlines

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3. Coherence, Gate Time, and Logical Depth

3.1 Raw Operations Are Not Logical Depth

The most important correction in this paper is conceptual:

Raw coherence time does not equal executable logical algorithm depth.

For example:

```
Raw physical gate opportunities = T1 / gate_time

If T1 = 2 ms and 2Q gate time = 25 ns:
raw physical gate opportunities = 2,000,000 ns / 25 ns = 80,000
```

That number is not the same thing as:

- logical gate depth
- full algorithm depth
- fault-tolerant executable workload length

In practice, logical depth depends on:

- repeated error-correction cycles
- decoder latency
- leakage handling
- calibration stability
- state preparation and measurement overhead
- idling and synchronization across modules

3.2 Implication for QONTOS

The relevant device question is therefore not:

```
How many raw gates fit inside T1?
```

It is:

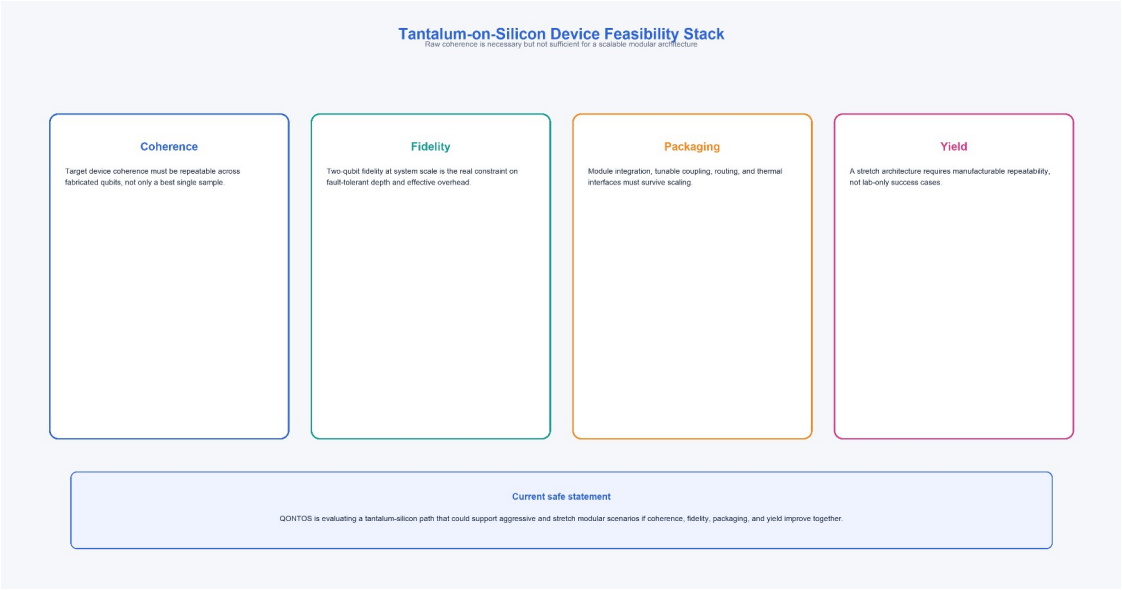
```
Does this device regime reduce the burden on fault tolerance enough
for the system-level architecture to remain plausible?
```

That is the right role for the tantalum platform in the QONTOS story.

3.3 Workload-Oriented Device Requirements

Workload class	Device implication
Early logical-qubit experiments	aggressive case may be sufficient
Modular fault-tolerance demonstrations	requires aggressive to stretch device regime
FeMoco-class flagship workloads	requires stretch device regime plus strong QEC and interconnect assumptions

4. Device Architecture and Control



Device Feasibility Stack

Figure 2: Raw coherence matters, but scalable device feasibility depends on fidelity, packaging, and yield together.

4.1 Double-Transmon Coupler Targets

Parameter
Tunable coupling window

CZ gate fidelity

CZ gate time

ZZ suppression

4.2 What Must Be Validated

The coupler architecture must be evaluated not only for isolated gate performance, but for:

- frequency crowding
- calibration drift
- cross-talk at dense packing
- leakage under repeated operation
- compatibility with chiplet packaging

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5. Fabrication and Packaging Strategy

5.1 Chiplet Assumptions

The current canonical stretch architecture assumes:

- 2,000 physical qubits per chiplet
- 5 chiplets per module
- 10,000 physical qubits per module

These are architecture constants, not demonstrated manufacturing outputs.

5.2 Fabrication Questions That Matter More Than Headlines

Question	Why it matters
Can yield remain acceptable at chiplet scale?	stretch architecture fails if yield collapses
Can packaging preserve qubit performance?	good bare-die metrics do not guarantee module-level success
Can control density remain manageable?	wiring and control are often harder than the qubits themselves
Can calibration labor be reduced enough?	manual tuning does not scale to large chiplet fleets

5.3 Partner Strategy

Named fabrication partners should be described as a partner strategy or target ecosystem, not as evidence that million-qubit production capacity has already been secured.

Recommended phrasing:

```
QONTOS is exploring a multi-partner fabrication strategy to reduce
supply-chain concentration and improve scale flexibility.
```

6. Thermal and Integration Constraints

6.1 Module Thermal Envelope

The current stretch planning envelope assumes roughly:

Component
Coaxial lines
Optical fibers
DC lines
Chip dissipation
Total
Available budget

This should be presented as a planning estimate tied to the cryogenic paper, not as an experimentally validated module budget.

6.2 Packaging and Calibration Risks

The device story is not only about physics. It is equally about:

- packaging yield
- thermal interfaces
- connector density
- calibration automation
- module replacement and maintainability

If these remain unresolved, even excellent single-device metrics will not convert into a viable modular machine.

7. Device Roadmap by Scenario

Year	Base	Aggressive	Stretch
2025-2026	literature benchmarking, design studies, calibration tooling	small internal validation targets	first evidence supporting millisecond-class stretch path
2026-2027	device-package integration	module-scale integration targets	10,000 physical-qubit stretch module assumptions tested
2027-2028	robust modular	high-fidelity chiplet	architecture-scale

	hardware support	fleet	stretch device envelope tested
2029-2030	strong device basis for modular products	large modular fault-tolerance path	full stretch device assumptions required

8. Validation Gates

The materials and device story should not be considered strong enough for the stretch architecture until the following are satisfied:

1. coherence and fidelity are shown under realistic packaging conditions
2. chiplet-level control and calibration complexity are bounded
3. fabrication yield assumptions are grounded in a real partner process model
4. module thermal and control density assumptions survive systems review

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9. Conclusion

Tantalum-on-silicon remains a promising candidate platform for the QONTOS modular architecture, especially because it offers a credible path toward stronger coherence and better device quality than older superconducting stacks. But this paper should not present raw coherence as equivalent to fault-tolerant algorithm readiness.

The technically defensible conclusion is:

- literature results make tantalum worth pursuing
- aggressive modular hardware scenarios may be plausible if packaging and control are solved
- the full million-qubit stretch architecture requires substantial additional validation beyond device-level headline metrics

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References

[1] Representative literature on tantalum transmons and millisecond-class coherence.

[2] Representative literature on superconducting gate fidelity, packaging, and calibration limits.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Device-feasibility analysis, not demonstrated stretch performance

Toward Quantum Error Correction at 100:1 Effective Overhead

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Correspondence: research@zhyra.xyz

Abstract

This paper evaluates a QONTOS path toward low-overhead fault tolerance for modular quantum systems. It does not claim that QONTOS has already achieved **100:1 effective physical-to-logical overhead**. Instead, it compares three overhead regimes - conservative, aggressive, and stretch - and examines the device, decoder, and architecture conditions required for each. The focus is on whether a hybrid code strategy combining practical surface-code ideas with more efficient LDPC-inspired approaches could reduce the physical cost of logical qubits enough to make large modular systems economically and operationally plausible.

Claim status: feasibility-envelope paper with a stretch target of 100:1 effective overhead

Keywords: quantum error correction, LDPC, surface code, decoder latency, fault-tolerant computing

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1. Introduction

Error-correction overhead is one of the central variables controlling whether large-scale quantum computing is merely scientifically interesting or actually buildable. For QONTOS, overhead is not a standalone coding-theory curiosity. It is the variable that couples:

- architecture scale
- facility size
- power budget
- capex
- algorithm viability

1.1 Claim Status Summary

Claim	Status
1000:1 is a conservative large-scale planning baseline	Derived from literature
300:1 is an aggressive QONTOS target	QONTOS target

100:1 is the QONTOS stretch target	Stretch target
QONTOS has already achieved 100:1	Not supported

1.2 Why This Matters

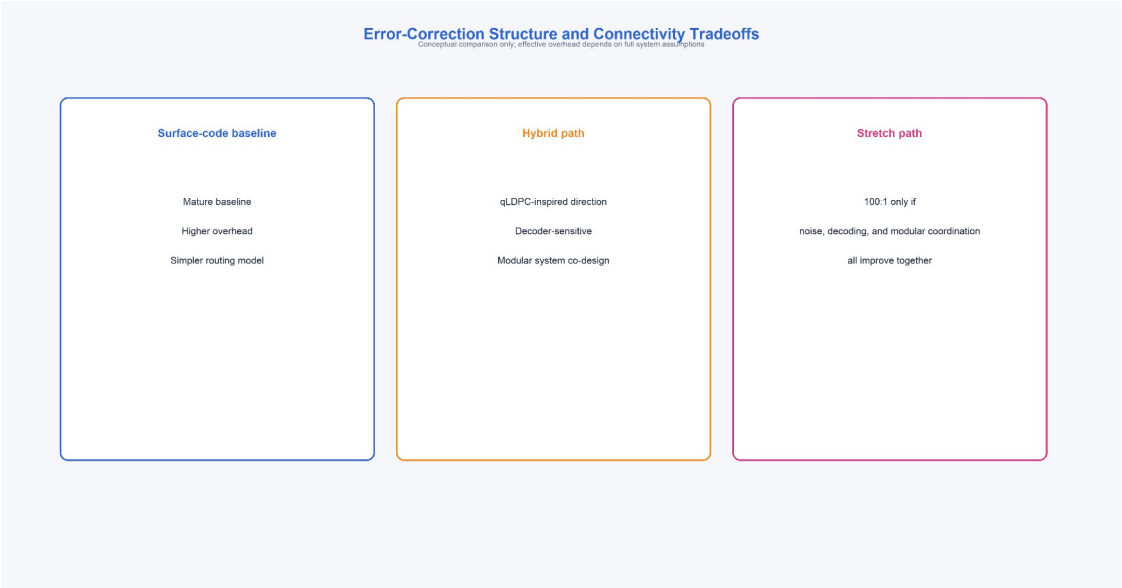
For 10,000 logical qubits:

- at 1000:1 overhead: 10,000,000 physical qubits
- at 300:1 overhead: 3,000,000 physical qubits
- at 100:1 overhead: 1,000,000 physical qubits

This is why the QEC story and the architecture story must stay synchronized.

---

2. Overhead Scenarios



Code Structure Tradeoffs

Figure 1: Surface-code baselines, hybrid-code paths, and stretch targets should be kept conceptually distinct.

2.1 Conservative, Aggressive, and Stretch Cases

Overhead regime
1000:1
300:1
100:1

2.2 Hybrid Code Framing

QONTOS is evaluating a hybrid direction:

- surface-code-like strategies where they remain operationally practical
- LDPC-inspired or more efficient coding strategies where connectivity and decoder assumptions permit
- architecture-aware code selection for modular hardware

This should be described as a design exploration, not as a settled production code stack.

2.3 Architecture Mapping

Using the canonical stretch architecture:

Layer
Chiplet
Module
System
Data center

The paper set must keep these counts aligned everywhere.

---

3. Decoder and Connectivity Requirements

3.1 What 100:1 Would Require

The stretch 100:1 case is not only a code question. It likely also requires:

- lower physical error rates than conservative baselines
- manageable 6-way or similar connectivity
- decoder latency near or below the correction-cycle budget
- tolerance to correlated noise and leakage
- modular operation without excessive communication penalties

3.2 Scenario Table

Variable
Effective overhead
Decoder latency
Decoder accuracy
Physical error
Connectivity burden

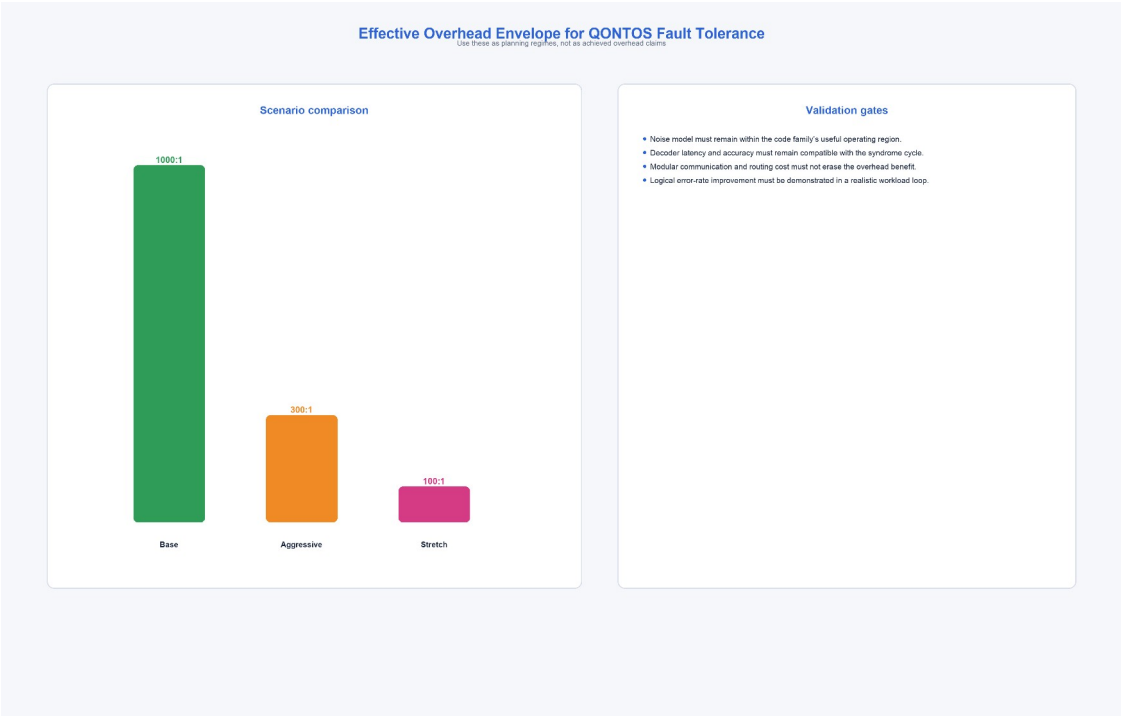
3.3 Decoder Performance Interpretation

The decoder table should not imply that `<1 us` or `500 ns` is already achieved by QONTOS. The correct framing is:

- `Base`: classical feasibility
- `Aggressive`: engineering target for modular logical qubits
- `Stretch`: required for the 100:1 architecture case

---

4. Implementation Strategy



Overhead Envelope

Figure 2: Effective overhead should be communicated as a planning envelope with explicit validation gates.

4.1 Syndrome Extraction

At large scale, the decoder and control path is a systems problem:

- syndrome extraction rate
- classical transport bandwidth
- local and global correction timing
- interaction with modular communication

Any low-overhead code strategy that ignores these operational costs is incomplete.

4.2 AI-Assisted Decoding

QONTOS should describe AI-assisted decoding as one possible enabler of the aggressive and stretch cases, not as a proof that those cases are already unlocked.

Recommended wording:

```
QONTOS evaluates AI-assisted decoding as a potential route to reducing
practical decoding latency and improving the viability of lower-
overhead code families in modular settings.
```

---

5. Fault-Tolerance Envelope

5.1 Threshold Framing

It is reasonable to discuss threshold and margin conceptually, but this paper should avoid implying that a single threshold number settles the architecture.

Better framing:

- physical error rate and threshold margin are necessary but not sufficient
- realistic performance also depends on leakage, correlated noise, finite-size effects, and decoder behavior

5.2 Logical Error Framing

The previous draft overstated logical performance through simplified asymptotic expressions. The revised paper should say:

```
Simplified asymptotic scaling suggests that sufficiently low physical
error and sufficient code distance could drive strong logical
suppression. However, the practical logical error rate in a modular
architecture must be estimated through finite-size analysis, realistic
noise models, and decoder-in-the-loop simulation.
```

5.3 Risks and Failure Modes

Major failure modes include:

- overhead remains far above 300:1
- decoder latency dominates correction cycle time
- correlated noise invalidates simplified suppression assumptions
- modular communication overhead destroys the logical cost advantage

---

6. Validation Gates

The low-overhead claim should only be promoted as a serious QONTOS program result once the following are satisfied:

1. a derivation exists for base, aggressive, and stretch overhead cases
2. decoder-in-the-loop simulation supports those cases
3. modular architecture penalties are included
4. first logical-qubit experiments align with the modeled trajectory

6.1 Phase-Aligned Gates

Phase
FOUNDATION
SPUTNIK
PIONEER
HORIZON
SUMMIT

7. Conclusion

The value of the QONTOS error-correction thesis is real, but it must be communicated precisely. The most technically defensible position today is:

- low-overhead fault tolerance is central to the architecture
- 300:1 is a meaningful aggressive target
- 100:1 remains a stretch target whose feasibility depends on device, decoder, and modular-systems performance improving together

That framing keeps the ambition while making the production plan more technically honest.

References

[1] Representative literature on surface codes, LDPC approaches, and low-overhead fault tolerance.

[2] Representative literature on decoder latency, finite-size effects, and modular fault tolerance.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Feasibility envelope for aggressive and stretch QEC targets

Photonic Interconnects for Modular Quantum Systems: From Literature Baseline to Stretch Architecture

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper evaluates a photonic-interconnect path for modular quantum systems in the QONTOS architecture. It does not claim that QONTOS has already achieved high-efficiency microwave-optical transduction or large-scale Bell-pair distribution. Instead, it compares literature baselines, aggressive engineering targets, and stretch architecture requirements for inter-module quantum communication. The paper treats photonic interconnects as a key scaling thesis because they may materially reduce wiring and thermal bottlenecks relative to purely electrical modular coupling, provided that efficiency, fidelity, and entanglement-rate thresholds are met.

Claim status: literature-informed systems feasibility paper with aggressive and stretch interconnect targets

---

1. Why Photonic Interconnects Matter

Modular quantum computing only becomes compelling if module-to-module communication is strong enough that scaling benefits outweigh communication penalties. For QONTOS, photonic interconnects are attractive because they may:

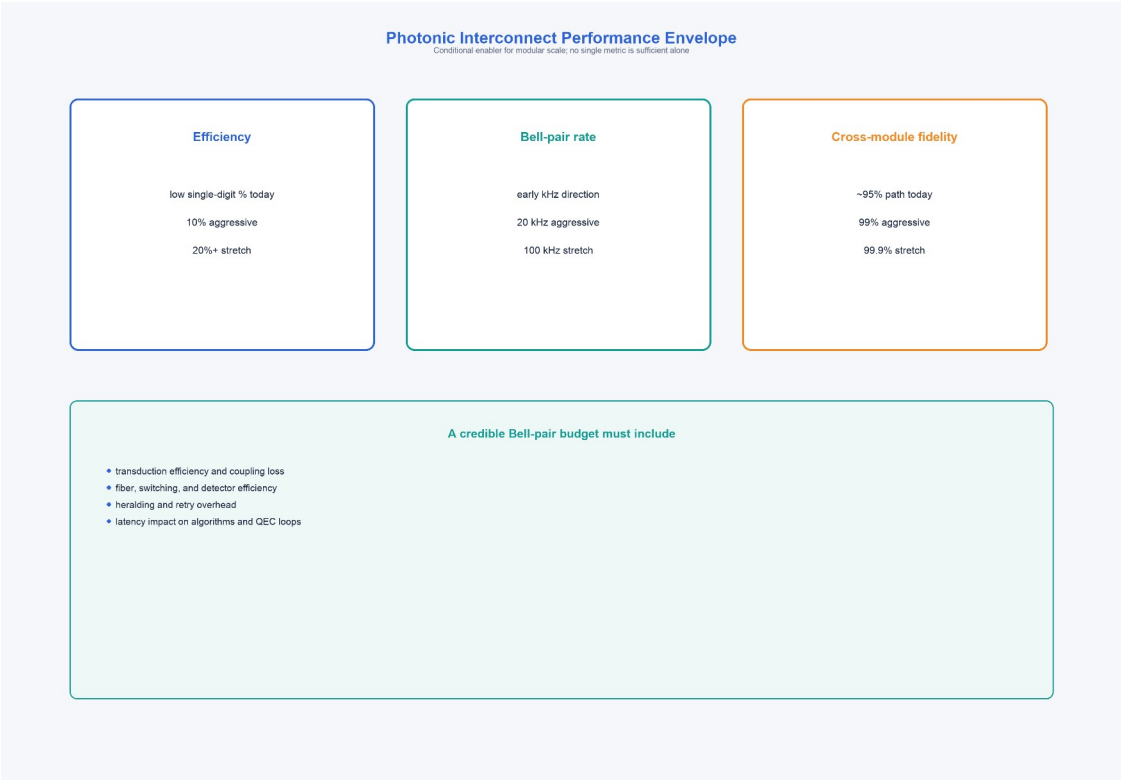
- reduce inter-module thermal load
- extend communication distance beyond local cryogenic packaging
- support modular scaling without monolithic chip growth

1.1 Claim Status Summary

Claim	Status
photonic links are a credible modular scaling direction	Derived from literature
QONTOS has achieved >20% transduction efficiency	Not supported
>20% transduction is a stretch requirement for the stretch architecture	Stretch target

large-scale Bell-pair distribution is central to the modular roadmap	QONTOS target

2. Scenario-Based Performance Envelope



Photonic Performance Envelope

Figure 1: Interconnect performance should be framed as a multi-metric envelope, not a single achieved efficiency headline.

2.1 Core Interconnect Targets

Metric
Transduction efficiency
Bell-pair rate
Cross-module fidelity
End-to-end latency

2.2 Literature Baseline vs QONTOS Target

Technology family	Literature-level direction	Role in QONTOS
Electro-optic	low-single-digit efficiency regime	early baseline / R&D path
Piezo-optomechanical	higher upside if integrated successfully	aggressive to stretch target path

The correct QONTOS posture is:

- literature indicates photonic modular links are worth pursuing
- aggressive and stretch targets require substantial additional integration success

3. Architecture Mapping

The canonical stretch deployment assumed in the paper set is:

- 5 chiplets per module
- 10 modules per system
- 10 systems per data center
- 100 modules total in the million-qubit stretch case

3.1 Interconnect Scale Implications

For the stretch architecture, the exact number of optical links depends on topology, redundancy, and whether counting is logical, directional, or physical. This paper should therefore avoid giving one unqualified number unless the topology is explicitly defined.

Safer phrasing:

```
Stretch-scale modular systems likely require hundreds to low thousands
of managed optical links, depending on topology and redundancy
assumptions.
```

3.2 Topology Options

Topology	Benefit	Risk
all-to-all within a system	lower routing overhead	optical complexity rises quickly
mesh between systems	bounded degree	extra hops and routing logic
staged hierarchy	operationally cleaner	increased scheduling complexity

4. End-to-End Bell-Pair Budget

4.1 Required Link-Budget Components

An honest photonic paper needs an explicit budget, not just headline rates.

The end-to-end Bell-pair story should include:

- transduction efficiency
- coupling loss
- fiber loss
- switch loss

- detector efficiency
- heralding rate
- resulting Bell-pair fidelity

4.2 What This Means for QONTOS

The stretch architecture is not supported by transduction efficiency alone. It requires:

- strong enough efficiency
- strong enough fidelity
- manageable retry overhead
- acceptable latency for distributed algorithms and QEC workflows

---

5. Failure Modes and Fallback Modes

5.1 Major failure modes

- transduction efficiency remains too low
- Bell-pair rate remains too low
- fidelity is too low for useful modular operations
- optical control complexity outweighs thermal advantages

5.2 Fallback modes

If the stretch photonic path underperforms, QONTOS can still retain a modular thesis by:

- reducing system size targets
- emphasizing shorter-range modularity first
- using software and digital-twin leadership while interconnect R&D matures

---

6. Validation Gates

The photonic thesis should only be promoted aggressively if the following gates are passed:

1. a two-module link budget is quantitatively defined
2. a credible Bell-pair generation experiment plan exists
3. modular runtime and algorithm studies include real communication penalties
4. thermal advantage over purely electrical scaling is clearly shown

---

7. Conclusion

Photonic interconnects remain one of the most important scaling ideas in the QONTOS research stack. But they should be described as a conditional enabler, not as a completed breakthrough. The strongest technically feasible framing is:

- photonic interconnects are a plausible path to modular scale
- aggressive and stretch performance levels remain target regimes
- the architecture only benefits if link efficiency, fidelity, and rate all improve together

---

References

[1] Representative literature on microwave-optical transduction and modular quantum networking.

[2] Representative literature on optical loss budgets, entanglement distribution, and modular superconducting architectures.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Interconnect feasibility analysis with explicit scenario framing

AI-Assisted Decoding for Modular Fault-Tolerant Quantum Systems

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper evaluates AI-assisted decoding as a possible enabler of lower-overhead fault tolerance in the QONTOS architecture. It does not claim that QONTOS has already achieved sub-microsecond decoder performance at million-qubit scale. Instead, it defines a latency budget, deployment assumptions, and validation plan for a hybrid decoding stack combining classical decoding, neural refinement, and accelerator-assisted correction. The decoder is treated as one of the key system bottlenecks linking error correction, architecture scale, and control infrastructure.

Claim status: target architecture paper with aggressive and stretch decoder assumptions

---

1. The Decoding Problem

At large scale, decoding becomes a full systems-engineering problem involving:

- syndrome extraction rate
- transport bandwidth
- memory pressure
- inference latency
- correction application timing
- modular synchronization

1.1 Claim Status Summary

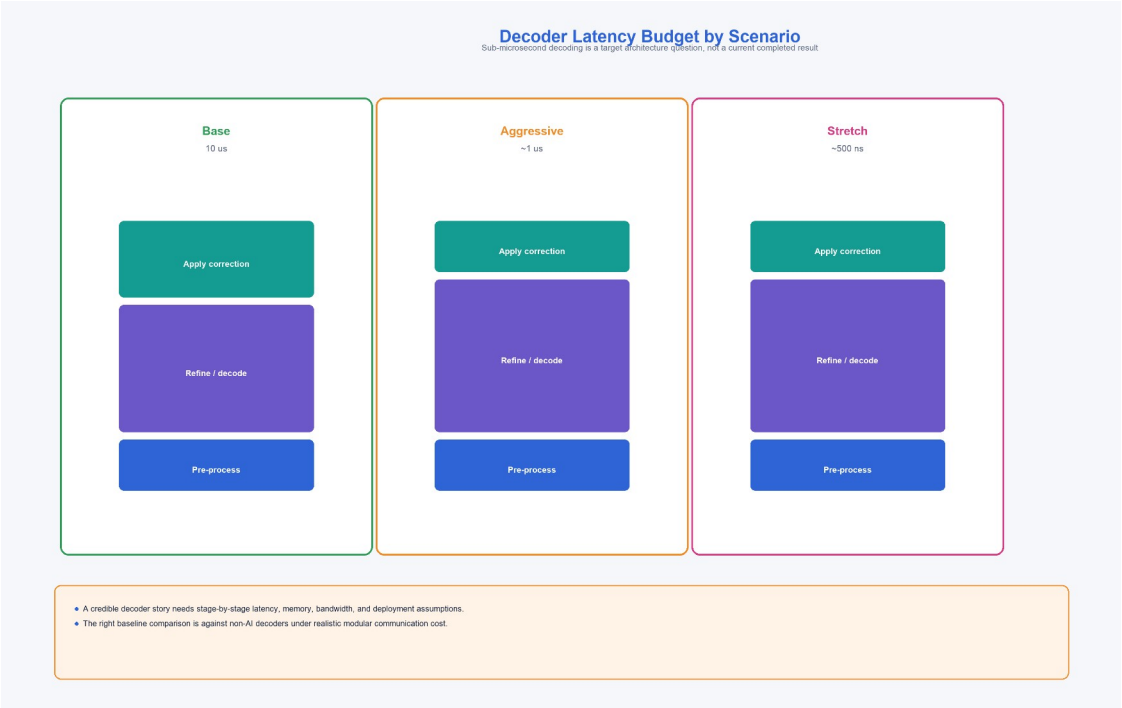
Claim	Status
AI-assisted decoding is a credible direction for scalable QEC	Derived from literature / QONTOS target
QONTOS has achieved <500 ns decoding latency	Not supported
<500 ns is a stretch deployment target	Stretch target

1.2 Scenario-Based Decoder Targets

Metric	Base	Aggressive	Stretch

Decoder latency	10 us	1 us	500 ns
Decoder accuracy	95%	99%	99.99%
Deployment mode	CPU/GPU hybrid	FPGA-assisted	FPGA or ASIC-class stretch path

2. Latency Budget



Decoder Latency Budget

Figure 1: The decoder story is strongest when presented as a staged latency budget rather than a declared solved result.

2.1 Target Pipeline Budget

The previous draft presented a sub-500 ns pipeline as if it were already achieved. The more defensible framing is a target latency budget:

Stage
Classical pre-processing
Ambiguity resolution / refinement
Correction application
Total

2.2 Interpretation

This budget should be read as:

- Base: plausibly achievable software-plus-accelerator regime
- Aggressive: serious engineering target for modular fault tolerance
- Stretch: deployment target requiring highly optimized accelerator paths

---

3. Decoder Stack

3.1 Functional Components

QONTOS should frame the decoder as a hybrid stack, not a single AI model:

- classical front-end decoding
- neural or learned refinement for hard cases
- accelerator-assisted correction dispatch

3.2 Why AI Matters

AI is most useful where:

- ambiguity is high
- classical heuristics saturate
- latency-pressure and accuracy-pressure coexist

It should not be presented as a magic replacement for all classical decoding.

---

4. Training, Inference, and Hardware Assumptions

4.1 Training Assumptions

The paper should state clearly that large training datasets are assumed, not yet demonstrated by QONTOS.

4.2 Inference Assumptions

If the decoder targets FPGA deployment, the paper must make explicit:

- model size assumptions
- memory assumptions
- quantization assumptions
- input/output bandwidth assumptions
- module-level power envelope

4.3 Power and Deployment Envelope

Recommended framing:

```
QONTOS treats low-latency decoder deployment as a co-design problem
spanning code choice, model choice, and hardware implementation.
```

---

5. System Risks

Major decoder-related risks include:

- model size too large for low-latency deployment
- bandwidth limits dominate inference time
- correlation structure differs from training assumptions
- distributed coordination costs erase local latency gains

---

6. Validation Gates

The decoder story should only be promoted aggressively once the following exist:

1. a full latency budget by stage
2. a hardware-in-the-loop or realistic accelerator benchmark
3. a clear baseline comparison against non-AI decoding
4. a modular-system correction-loop analysis including communication delays

---

7. Conclusion

AI-assisted decoding remains a potentially important enabler of lower-overhead modular fault tolerance in QONTOS. The technically feasible way to present it is:

- as a targeted decoder architecture
- as one component in a larger systems budget
- as a base/aggressive/stretch ladder rather than a declared solved result

---

References

[1] Representative literature on neural decoders, hybrid decoders, and accelerator-assisted QEC.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Decoder target architecture with explicit latency assumptions

Distributed Quantum Software Stack: From Current QONTOS Platform to Future Modular Operating Layer

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper describes the target QONTOS software stack for modular quantum systems while explicitly distinguishing current implemented capability from future scale ambitions. QONTOS already has a software-first orchestration platform centered on circuit ingestion, partitioning, scheduling, execution, aggregation, and observability. This paper extends that foundation into a future modular operating layer intended for larger distributed systems, hardware integration, and fault-tolerant workflows. The paper therefore serves as the bridge between the current implemented platform and the larger research architecture.

Claim status: mixed paper covering both implemented software capability and future target stack

---

1. Current and Future Roles of the Stack

QONTOS is at its strongest today as a software-first modular orchestration effort. That is an advantage, not a weakness, and this paper should say so clearly.

1.1 Current Implemented Capability

The current platform already supports a software-oriented control plane built around:

- circuit ingestion and normalization
- partitioning and scheduling
- simulator-backed execution paths
- result aggregation
- replayability and observability

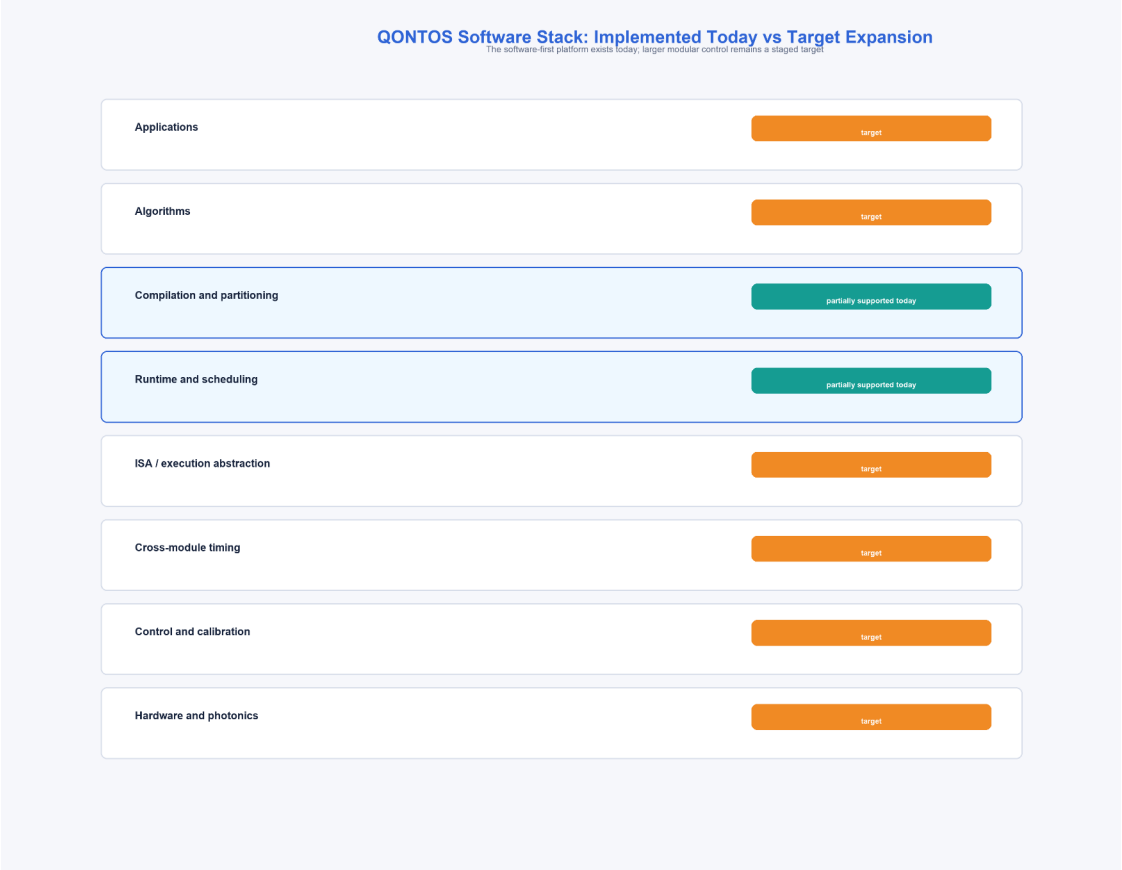
This is consistent with the implementation-grounded current whitepaper and should remain the public baseline.

1.2 Future Target Capability

The future software stack aims to extend that platform toward:

- modular hardware integration
- cross-module timing and control
- digital-twin-assisted scheduling
- fault-tolerant workloads
- larger logical-qubit-scale runtime management

2. Target Eight-Layer Architecture



Software Stack Layers

Figure 1: The QONTOS software stack should distinguish implemented layers from target expansion layers.

Layer
8
7
6
5
4
3

2

1

2.1 Important Framing Rule

The stack should not be described as already supporting million-qubit programming. The correct wording is:

```
QONTOS is developing a software stack intended to support large-scale
modular quantum systems, built on an implementation-grounded
orchestration platform that exists today.
```

---

3. Modular Runtime Functions

3.1 Present-day strengths

- orchestration model
- partition-aware workflow
- result aggregation and provenance
- benchmark and replayability foundation

3.2 Future modular functions

- module-aware circuit placement
- communication-aware scheduling
- dynamic calibration integration
- fault-tolerant workload control

---

4. Circuit Cutting and Modular Execution

Circuit cutting and modular reconstruction should be framed carefully.

Recommended posture:

- current: simulator-backed and methodology-oriented
- future: integrated into the modular runtime and digital twin

Do not imply that current QONTOS software already proves million-qubit modular execution in practice.

---

5. AI-Powered Optimization

The optimization layer should be treated as a target capability with staged goals:

Function

gate reduction
topology-aware mapping
predictive calibration support

These should remain target ranges unless benchmarked on the current platform.

---

6. Validation Gates

The software paper should define its own credibility gates:

1. current platform behavior is reproducible and benchmarked
2. digital twin is aligned with modular architecture assumptions
3. distributed runtime logic is tested against realistic modular workloads
4. hardware integration claims are kept separate from simulator-only capability

---

7. Conclusion

This paper should function as the bridge between present QONTOS and future QONTOS. The most important message is:

- QONTOS already has a real software-first foundation
- that foundation is the right base from which to pursue modular hardware integration
- large-scale modular software capability should be described as a target extension, not a completed state

---

References

[1] Current QONTOS implementation and orchestration whitepaper.

[2] Representative literature on distributed quantum runtime design and modular execution.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Implemented platform plus target modular software stack

Cryogenic Infrastructure for Large-Scale Modular Quantum Systems

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper outlines a cryogenic infrastructure concept for large-scale modular quantum systems in the QONTOS program. It does not claim that QONTOS has already demonstrated million-qubit cryogenic deployment. Instead, it evaluates the thermal and operational assumptions required for aggressive and stretch scenarios, including cooling power per module, wiring burden, control-electronics distribution, uptime, and facility-scale operation. The central thesis is that cryogenic feasibility depends not only on raw cooling numbers, but on packaging, control density, maintenance, redundancy, and the degree to which photonic interconnects reduce thermal bottlenecks.

Claim status: infrastructure planning paper with scenario-based thermal assumptions

---

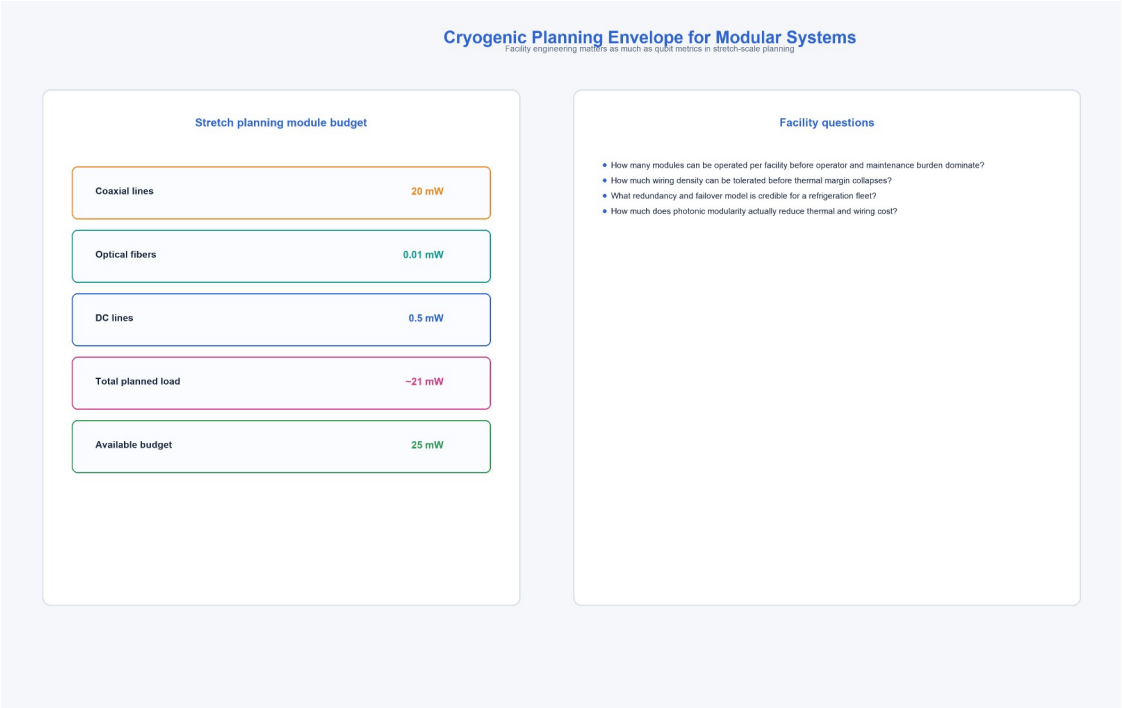
1. Why Cryogenics Is a First-Order Constraint

Cryogenic infrastructure is one of the main reasons modular quantum architectures must be treated as full systems problems rather than simple qubit-scaling exercises.

1.1 Claim Status Summary

Claim	Status
cryogenics is a major scaling constraint	Demonstrated engineering reality
25 mW/module at 15 mK is sufficient for stretch modules	Stretch planning assumption
100-module facility operation is feasible	Stretch target

2. Scenario-Based Thermal Envelope



Cryogenic Planning Envelope

Figure 1: Cryogenic feasibility should be presented as a module and facility planning problem, not as a solved deployment fact.

Metric
Cooling per module at 15 mK
Modules per facility
Facility power
Uptime

2.1 Module-Level Planning Budget

Component
Coaxial lines
Optical fibers
DC lines
Total
Available budget

This should be presented as a planning budget linked to architecture assumptions, not a demonstrated module result.

3. Wiring and Control Assumptions

Cryogenic feasibility depends on more than QPU heat load. It also depends on:

- control line count
- attenuation strategy
- amplifier placement
- electronics staging
- module serviceability

These should be explicit in this paper, because they often dominate whether a module concept is actually buildable.

---

4. Facility-Scale Questions

4.1 Stretch-Scale Facility Questions

The stretch architecture implies a facility-scale operating model. This creates real questions about:

- refrigerator fleet management
- maintenance windows
- redundancy and failover
- operator burden
- room-level infrastructure and power conditioning

4.2 Why Photonics Matters Here

Photonic interconnects matter to this paper because they may reduce some thermal and wiring burdens relative to purely electrical inter-module coupling. That benefit should be described as conditional and architecture-dependent, not automatic.

---

5. Failure Modes

Major cryogenic and facility risks include:

- wiring density grows faster than thermal margin
- module control overhead dominates the cryogenic budget
- maintenance burden reduces uptime below economic thresholds
- facility-scale orchestration and redundancy become operationally expensive

---

6. Validation Gates

The cryogenic thesis should only be promoted aggressively if the following exist:

1. a full wiring and attenuation budget
2. a module control-electronics architecture
3. a refrigeration fleet model for aggressive and stretch scenarios
4. a maintenance and uptime model with realistic redundancy assumptions

---

7. Conclusion

Cryogenic infrastructure is one of the strongest reasons to use scenario-based planning rather than headline-only scaling claims. The technically honest position is:

- modular cryogenic scale may be feasible
- photonic interconnects may help materially
- the million-qubit stretch case depends on facility engineering, not just qubit metrics

---

References

[1] Representative literature on dilution refrigeration, cryogenic control, and scaling limits in superconducting systems.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Scenario-based cryogenic planning and risk analysis

Quantum Algorithm Roadmap for Modular Fault-Tolerant Systems

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper defines the QONTOS algorithm roadmap for modular fault-tolerant quantum systems. It does not claim that QONTOS has already demonstrated readiness for FeMoco-scale computation. Instead, it establishes a benchmark ladder from small chemistry and optimization problems toward a stretch-case FeMoco flagship target. The paper treats FeMoco as a long-term benchmark because it is scientifically meaningful and commercially resonant, while emphasizing that algorithmic readiness must be built progressively through increasingly demanding workload classes.

Claim status: algorithm roadmap paper with benchmark-ladder framing

---

1. Why Algorithms Must Be Framed as a Ladder

Large-scale algorithm claims are only technically credible if they sit on top of smaller validated workloads. QONTOS should therefore avoid presenting FeMoco as an isolated endpoint without an intermediate progression.

1.1 Claim Status Summary

Claim	Status
FeMoco is a strong flagship benchmark	Derived from literature and strategy
QONTOS is ready for FeMoco-scale execution	Not supported
FeMoco remains the stretch benchmark for the 2030 roadmap	Stretch target

2. Benchmark Ladder

Algorithm Benchmark Ladder

Figure 1: FeMoco is the stretch flagship, but credibility comes from a ladder of smaller validated workloads.

2.1 Chemistry Ladder

Stage	Workload	Role
C1	H2 / LiH class systems	near-term benchmark harness and software validation
C2	small active-space chemistry	digital-twin and modular runtime progression
C3	medium catalyst models	aggressive modular benchmark
C4	FeMoco-class problem	stretch flagship benchmark

2.2 Optimization Ladder

Stage	Workload	Role
O1	small QAOA or routing instances	present software and scheduler validation
O2	medium portfolio and logistics instances	digital-twin and modular progression
O3	large distributed	aggressive architecture

optimization cases target

2.3 Machine-Learning Ladder

Stage
M1
M2
M3

3. FeMoco as the Stretch Flagship

3.1 Why FeMoco

FeMoco remains attractive because it is:

- scientifically important
- widely recognized as a quantum benchmark
- linked to compelling industrial narratives

3.2 Correct Framing

FeMoco should be described as:

- a flagship stretch benchmark
- a long-term target
- an anchor for resource-estimation and benchmark methodology

It should not be described as already enabled by the current QONTOS stack.

3.3 Resource-Estimation Posture

Any qubit, gate-count, runtime, or error-rate table should be explicitly labeled as:

- literature-derived
- simulator-estimated
- digital-twin-estimated
- QONTOS target

4. Scenario-Based Algorithm Outlook

Scenario	What is technically realistic to discuss
Base	small chemistry and optimization benchmarks on the current platform and digital twin
Aggressive	medium modular workloads and early fault-tolerant benchmark progression

Stretch	FeMoco-class or similarly demanding flagship workloads

4.1 Safe Public Message

```
QONTOS uses FeMoco as a long-term benchmark target while progressing
through a staged ladder of smaller chemistry, optimization, and
modular benchmark workloads.
```

---

5. Validation Gates

The algorithm story should only move upward when these are satisfied:

1. smaller benchmark classes run reproducibly
2. resource-estimation methodology is clearly documented
3. benchmark ladder stages are tied to architecture and QEC assumptions
4. flagship algorithm claims are only made after intermediate milestones are passed

---

6. Conclusion

This paper is strongest when it treats algorithms as the final proof layer of the QONTOS architecture rather than as a standalone vision statement. The technically feasible narrative is:

- QONTOS has a benchmark ladder
- FeMoco remains the stretch flagship
- progress should be measured through staged workload classes, not one heroic endpoint

---

References

[1] Representative literature on FeMoco, fault-tolerant chemistry, and algorithmic resource estimation.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Benchmark-ladder algorithm roadmap

Benchmarking Framework for Large-Scale Modular Quantum Systems

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper defines the benchmarking framework that QONTOS should use to measure progress toward large-scale modular quantum computing. It does not present million-qubit benchmark values as measured QONTOS results. Instead, it establishes benchmark categories, measurement conditions, scenario-based targets, and validation gates spanning simulator-backed orchestration, digital-twin modeling, hardware integration, and eventual fault-tolerant modular systems. The framework includes system metrics, logical-qubit metrics, and application-level benchmarks, with FeMoco retained as a flagship stretch benchmark rather than the sole validation target.

Claim status: methodology paper for benchmark design and release gating

---

1. Why Benchmarking Must Lead the Narrative

The QONTOS research program will only remain credible if every major architecture and roadmap claim can be mapped to a benchmark, protocol, or validation gate.

This paper therefore serves a special role:

- it defines how progress will be measured
- it distinguishes measured values from targets
- it tells readers how to compare current QONTOS capability with future scenarios

1.1 Claim Status Summary

Claim	Status
QONTOS has a current simulator-backed benchmark capability	Supported internally
QONTOS has measured million-qubit benchmark results	Not supported
QONTOS has stretch benchmark targets for million-qubit systems	QONTOS target / stretch target

1.2 Benchmark Categories

QONTOS should report benchmarks in four categories:

Category	Meaning
Implemented platform	Measured on current software stack and benchmark harness
Digital twin	Simulated in modular hardware and runtime models
Hardware integration	Measured on real devices or module prototypes
Stretch architecture	Scenario target under explicit assumptions

2. Measurement Conditions

Every reported benchmark should identify:

- claim label
- environment
- workload
- noise model
- hardware assumptions
- reconstruction or approximation method
- confidence interval or uncertainty note

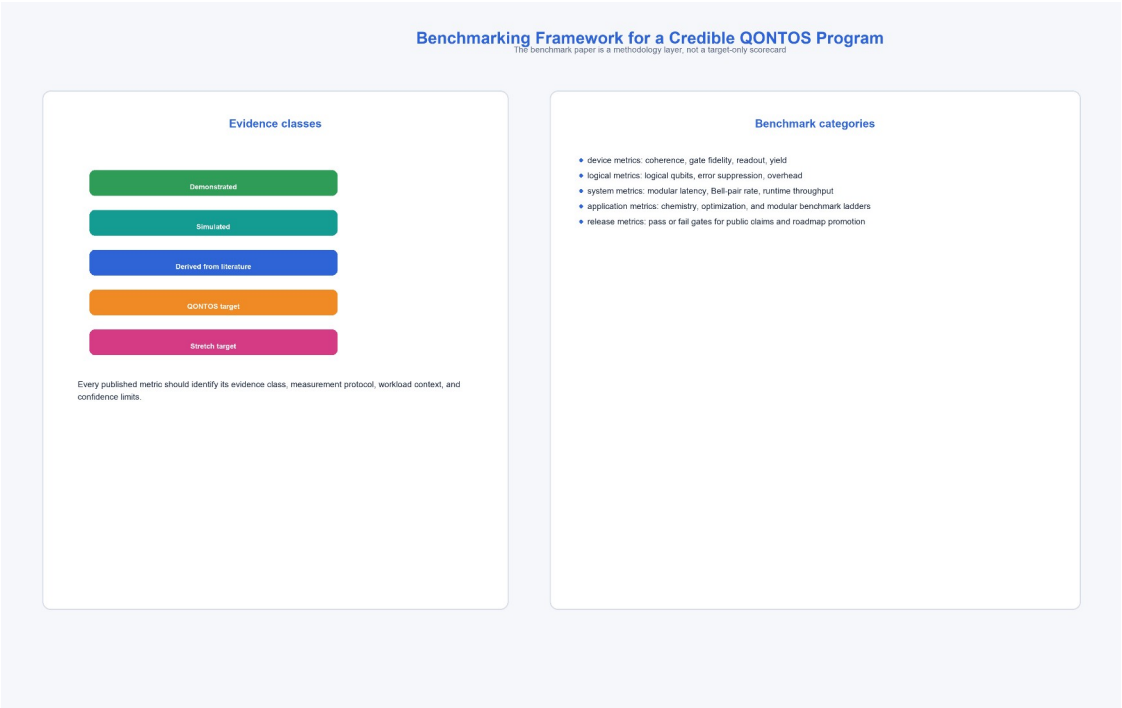
2.1 Mandatory Metadata for Every Result

Field
claim label
benchmark layer
simulator, digital twin, or hardware
workload size
noise or fidelity assumptions
partitioning / modular assumptions
pass/fail threshold

2.2 Benchmark Layers

Layer
L0
L1
L2
L3

3. System Benchmarks



Benchmarking Framework

Figure 1: QONTOS benchmarking should classify every metric by evidence level and benchmark type.

3.1 Metric Definitions

Metric	What it means	Safe use
Quantum Volume	system-level randomized benchmark	only if measurement protocol is explicit
CLOPS	circuit layer throughput	only if execution environment is explicit
Logical qubit count	simultaneously usable logical resources	must define what counts as logical
Logical error rate	post-correction error performance	must include code and decoder assumptions

3.2 Scenario Targets

Metric	Base	Aggressive	Stretch	Claim posture
Logical qubits	1-20	100-1,000	10,000	target range
Quantum	benchmark-	benchmark-	10^15	stretch

Volume	defined	defined		target
CLOPS	benchmark-defined	benchmark-defined	10^12	stretch target
Runtime-supported circuit depth	benchmark-defined	benchmark-defined	10^12-scale flagship class	stretch target

3.3 Competitive Comparison Rule

Do not mix:

- measured IBM or Google values
- QONTOS stretch targets

inside the same table without clearly separating:

- measured current values
- literature projections
- QONTOS targets

---

4. Logical-Qubit Metrics

4.1 Required Definitions

QONTOS should define a logical qubit as usable only if:

- the code family is specified
- the decoder path is specified
- the correction cycle budget is specified
- the logical error target is specified

4.2 Scenario-Based Logical Metrics

Metric	Base	Aggressive	Stretch
Effective overhead	1000:1	300:1	100:1
Logical error target	1e-4	1e-6	1e-8
Logical coherence goal	benchmark-defined	benchmark-defined	long-duration flagship workloads

4.3 What Not To Do

Do not report:

- logical qubit count
- logical coherence

- logical advantage

without stating the exact error-correction, decoder, and architecture assumptions.

---

5. Application Benchmark Ladder

QONTOS should benchmark progress using a staircase of workloads rather than jumping directly to FeMoco.

5.1 Chemistry Ladder

Stage
C1
C2
C3
C4

5.2 Optimization Ladder

Stage
O1
O2
O3

5.3 Flagship Benchmark Rule

FeMoco should remain the flagship stretch benchmark, but it must not be the only benchmark.

---

6. Current QONTOS Benchmark Surface

The current QONTOS benchmark story should stay tied to the implemented software-first platform:

- simulator-backed orchestration
- partitioning and scheduling experiments
- digital-twin projections
- result aggregation and replayability

This is the safe foundation from which future modular and hardware benchmarks can grow.

6.1 Current Safe Public Framing

```
QONTOS has a benchmark program for modular orchestration and digital-
twin validation today, and is defining a larger benchmark framework
for future hardware-integrated modular systems.
```

---

7. Pass/Fail Gates

Each benchmark family should have an explicit gate.

Gate	Required outcome
G1	benchmark protocol defined and reproducible
G2	current platform benchmarks run cleanly and are versioned
G3	digital-twin results can be reproduced from assumptions
G4	hardware results, when present, are clearly separated from targets
G5	flagship application claims are tied to a benchmark ladder, not a single heroic endpoint

8. Release and Communication Rules

8.1 Safe benchmark wording

- target
- stretch benchmark
- digital-twin projection
- simulator-backed result
- measured on current platform

8.2 Unsafe benchmark wording unless measured

- QONTOS achieves Quantum Volume 10^15
- QONTOS achieves CLOPS 10^12
- QONTOS has 10,000 logical qubits operational
- QONTOS demonstrates quantum advantage

---

9. Conclusion

Benchmarking should be the credibility engine of the QONTOS research and production program. The most important shift in this paper is from target declaration to measurement design. Once QONTOS can show how it defines, measures, and gates each major claim, the rest of the paper set becomes much more technically defensible.

References

[1] Representative literature on Quantum Volume, CLOPS, logical benchmarking, and application-level FTQC evaluation.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Benchmark methodology and gating framework

Roadmap to Large-Scale Modular Quantum Computing: Base, Aggressive, and Stretch Scenarios Through 2030

Technical Research Paper v2.1

Author: Qontos Research Wing

Affiliation: Zhyra Quantum Research Institute (ZQRI), Abu Dhabi, UAE

Abstract

This paper presents a gated development roadmap for QONTOS through 2030. It replaces a single-path million-qubit storyline with three scenarios: `Base`, `Aggressive`, and `Stretch`. The stretch scenario retains the long-term aim of **1,000,000 physical qubits** and **10,000 logical qubits**, but the paper now treats that outcome as conditional on explicit subsystem breakthroughs in device performance, error correction, interconnects, cryogenics, and software control. The purpose of the roadmap is therefore not to guarantee a result. It is to define a disciplined sequence of gates, fallback outcomes, and decision points.

Claim status: program roadmap and scenario planning document

---

1. Roadmap Philosophy

The previous roadmap framing was too singular. A technically feasible program needs:

- base outcome
- aggressive outcome
- stretch outcome
- validation gates
- no-go conditions
- fallback paths

1.1 Scenario Summary

Scenario	By 2030 likely outcome
Base	strong software, digital twin, modular hardware integration, early logical qubits
Aggressive	large modular machine with meaningful logical-qubit and distributed FTQC demonstrations
Stretch	1,000,000 physical qubits and 10,000 logical qubits

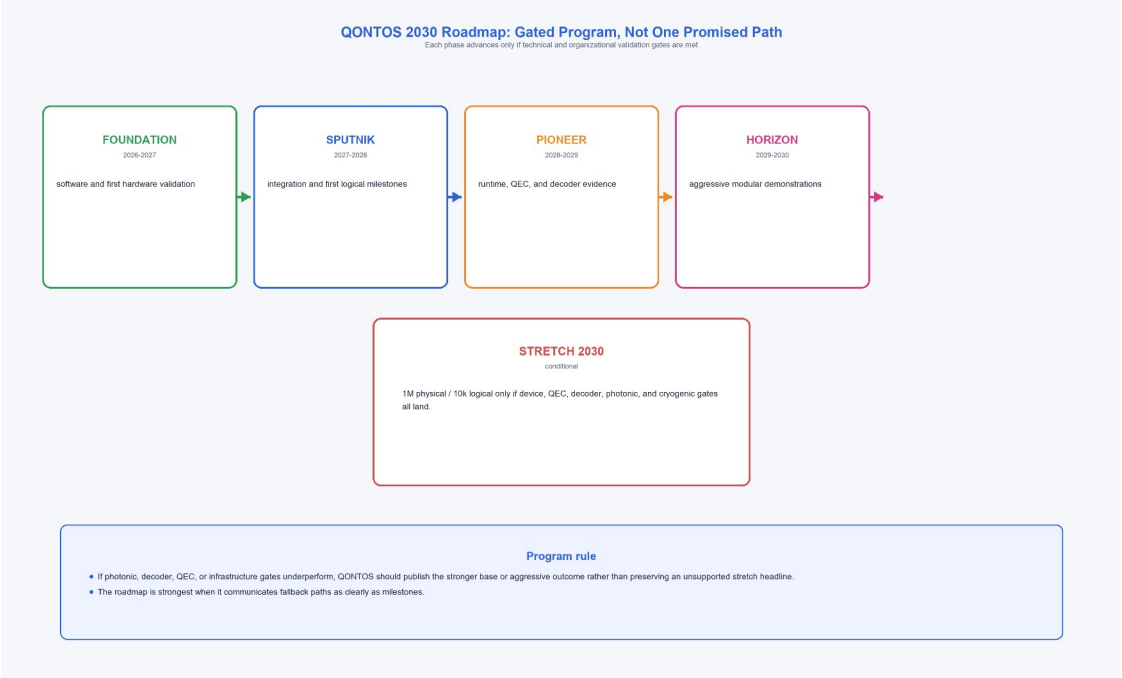
1.2 What This Roadmap Does Not Claim

This roadmap does not claim:

- that the stretch scenario is already validated
- that the stretch budget is guaranteed to be sufficient
- that all subsystem breakthroughs will land on the same timeline

---

2. Five Program Phases



Gated Roadmap

Figure 1: The roadmap should be read as a gated program with fallback outcomes, not as one promised path.

2.1 FOUNDATION (2025-2026)

Primary objective

- establish the software, benchmark, device, and digital-twin foundation

Base outcome

- strong software-first orchestration platform
- current benchmark harness and replayability
- improved architecture and assumptions discipline

Aggressive outcome

- first hardware-linked validation path

- stronger device and packaging evidence

Stretch relevance

- first evidence that stretch device and QEC assumptions are not obviously impossible

Validation gates

- canonical architecture constants fixed
- benchmark methodology paper in place
- first device assumptions table grounded in literature
- current platform and simulation artifacts reproducible

No-go conditions

- no coherent benchmark program
- no plausible device story
- no current platform credibility

Fallback if gate fails

- remain a software-first orchestration and digital-twin platform while hardware path matures

2.2 SPUTNIK (2026-2027)

Primary objective

- move from architecture thesis to first scaled modular hardware assumptions

Base outcome

- small modular hardware integration path
- digital-twin and software stack mature together

Aggressive outcome

- 10,000 physical-qubit class module assumptions become defensible
- first 100 logical-qubit program target becomes credible

Stretch relevance

- first evidence toward the 10,000 physical-qubit module target

Validation gates

- packaging and thermal budgets reviewed
- first logical-qubit milestone path defined
- inter-module entanglement experiment plan credible

No-go conditions

- module thermal/control assumptions fail
- device packaging does not preserve performance

Fallback if gate fails

- reduce module size and extend timeline while preserving modular software roadmap

2.3 PIONEER (2027-2028)

Primary objective

- validate multi-module behavior and distributed execution

Base outcome

- mature distributed runtime and modular benchmark program

Aggressive outcome

- large multi-module demonstrations
- 100-1,000 logical-qubit path becomes plausible

Stretch relevance

- first credible path to 100,000 physical qubits and modular FTQC subroutines

Validation gates

- distributed runtime and digital twin aligned
- inter-module communication passes minimum thresholds
- decoder path supports the targeted correction-cycle budget

No-go conditions

- communication overhead dominates
- decoder latency or reliability collapses at modular scale

Fallback if gate fails

- focus on lower-scale modular advantage and software commercialization

2.4 HORIZON (2028-2029)

Primary objective

- translate modular maturity into large-scale logical performance

Base outcome

- strong modular system platform with useful workload demonstrations

Aggressive outcome

- 100,000 physical-qubit and 100-1,000 logical-qubit class system becomes feasible

Stretch relevance

- 500,000 physical qubits and 5,000 logical qubits become conditionally plausible

Validation gates

- effective overhead moves materially below conservative baselines
- photonic and cryogenic assumptions survive systems validation
- benchmark ladder shows meaningful upward progression

No-go conditions

- effective overhead remains near 1000:1
- photonic path remains too weak for modular scaling

Fallback if gate fails

- keep aggressive system target, retire or defer the stretch case

2.5 SUMMIT (2029-2030)

Primary objective

- determine whether the stretch architecture is truly reachable

Base outcome

- commercially meaningful modular platform with validated logical resources

Aggressive outcome

- large fault-tolerant modular machine with strong flagship benchmark progress

Stretch outcome

- 1,000,000 physical qubits
- 10,000 logical qubits
- flagship chemistry or comparable application advantage

Validation gates

- device assumptions support stretch performance
- effective overhead near stretch case
- interconnect assumptions near stretch case
- benchmark and replay evidence support the claim set

No-go conditions

- one or more critical subsystems remain at base or conservative performance

Fallback if gate fails

- ship the strongest aggressive architecture achievable and continue stretch R&D beyond 2030

---

3. Scenario Tables

3.1 Architecture Outcome by 2030

Metric
Physical qubits
Logical qubits
Effective overhead
Interconnect regime

3.2 Investment Envelope

Scenario	Indicative interpretation
Base	software-led and hardware-integrating program
Aggressive	full modular research and scale-up program
Stretch	approximately the previously cited $530-780M class envelope, contingent on subsystem success

The stretch investment number should remain an estimated planning envelope, not a guarantee of sufficiency.

---

4. Risks and Dependencies

4.1 Major technical dependencies

- millisecond-class device regime
- low-enough effective overhead
- practical decoder latency
- adequate transduction and Bell-pair performance
- cryogenic and control scalability

4.2 Major program dependencies

- fabrication partners
- packaging and supply chain
- sustained capital access
- team build-out across software, hardware, cryogenics, and QEC

4.3 Major communication risk

The biggest communication risk is presenting the stretch scenario as if it were already a technically consolidated production plan. This paper should explicitly avoid that.

---

5. Recommended External Framing

The most technically credible public statement is:

```
QONTOS is pursuing a modular quantum roadmap with base, aggressive,
and stretch scenarios through 2030. The stretch scenario targets 1
million physical qubits and 10,000 logical qubits, contingent on major
advances in qubits, error correction, interconnects, cryogenics, and
control systems.
```

Avoid:

- `QONTOS will have 1 million qubits by 2030`
- `QONTOS will solve FeMoco by 2030`
- `The quantum future is 5 years away`

unless explicitly framed as stretch-case language.

---

6. Conclusion

This roadmap is strongest when it is used as a disciplined program map rather than a single-path promise. The stretch scenario remains valuable because it defines a bold north star. But the base and aggressive scenarios are equally important because they preserve technical credibility and create real decision points.

The roadmap is therefore feasible only if it remains gated, scenario-based, and tightly linked to benchmark evidence.

References

[1] Internal QONTOS architecture constants, assumptions register, and claim-status guidance.

[2] Representative literature on modular quantum systems, error correction, and large-scale superconducting roadmaps.

Document Version: 2.1
Classification: Technical Research Paper
Claim posture: Scenario-based roadmap with explicit gates and fallback paths